

PORTS: Preference-Optimized Retrievers for Tool Selection with Large Language Models

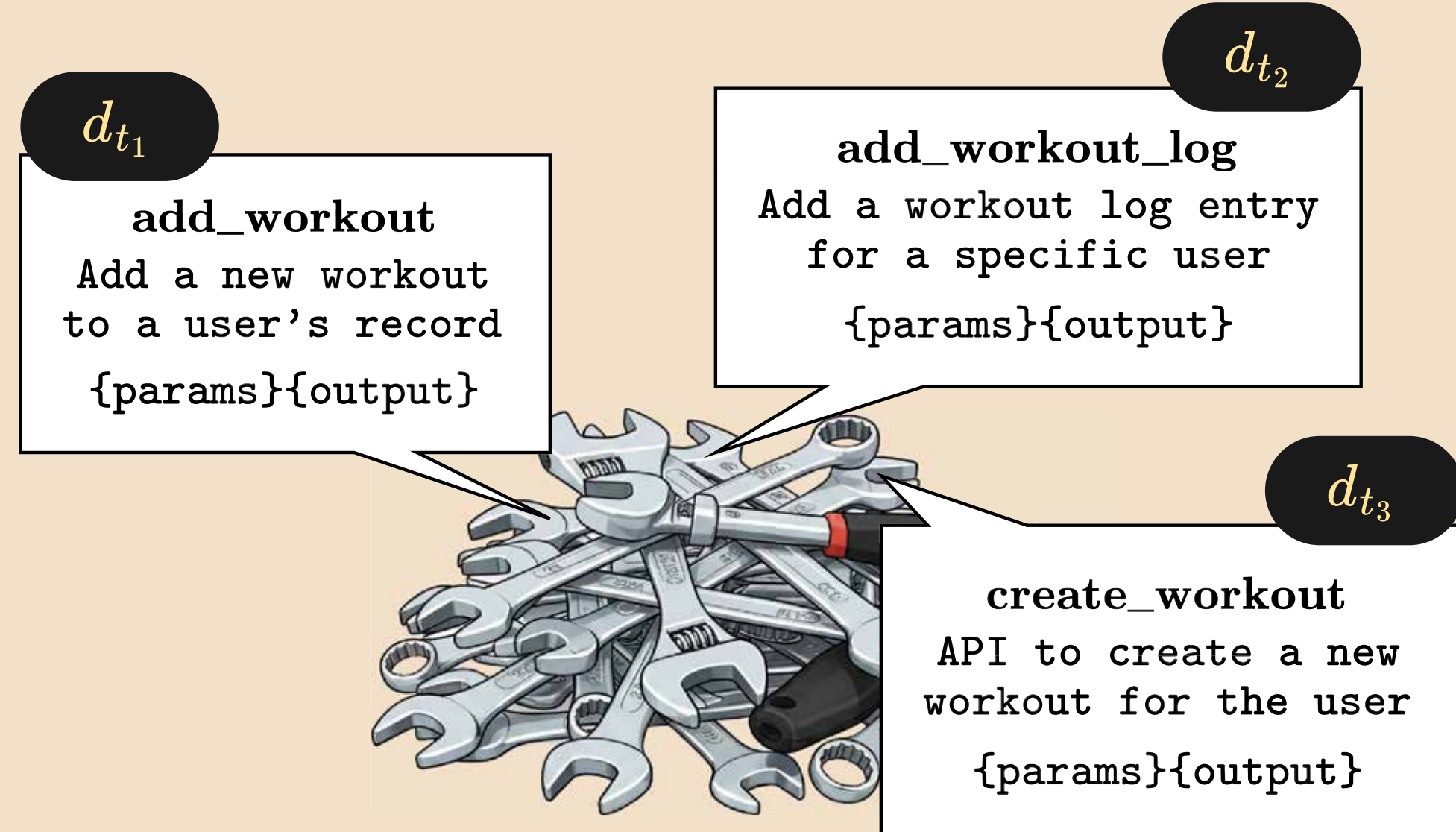
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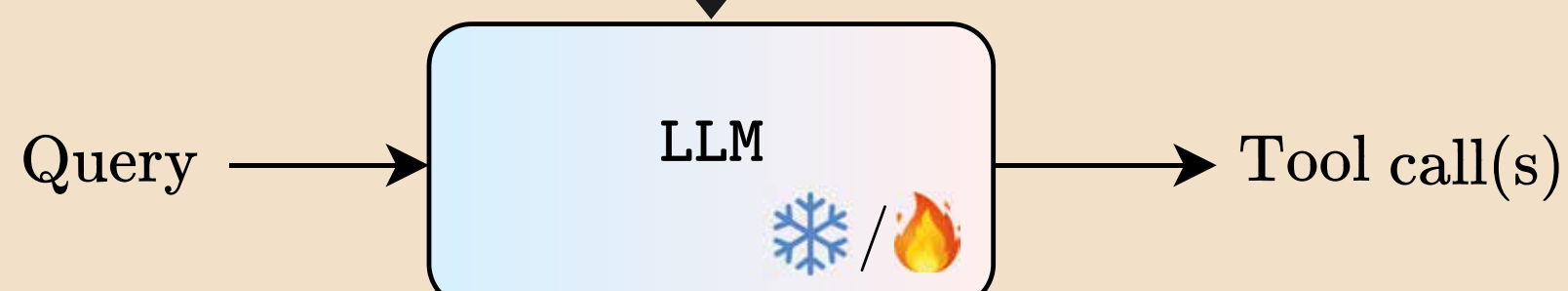
Motivations



The number of target tools N for agentic applications is increasing every day. In some datasets, they already exceed 100K.

You are a tool calling model. You are provided with tool docstrings within `<tools></tools>` XML tags. You may call one or more tools to assist with the user query.

`<tools> dt1 ... dtN </tools>`

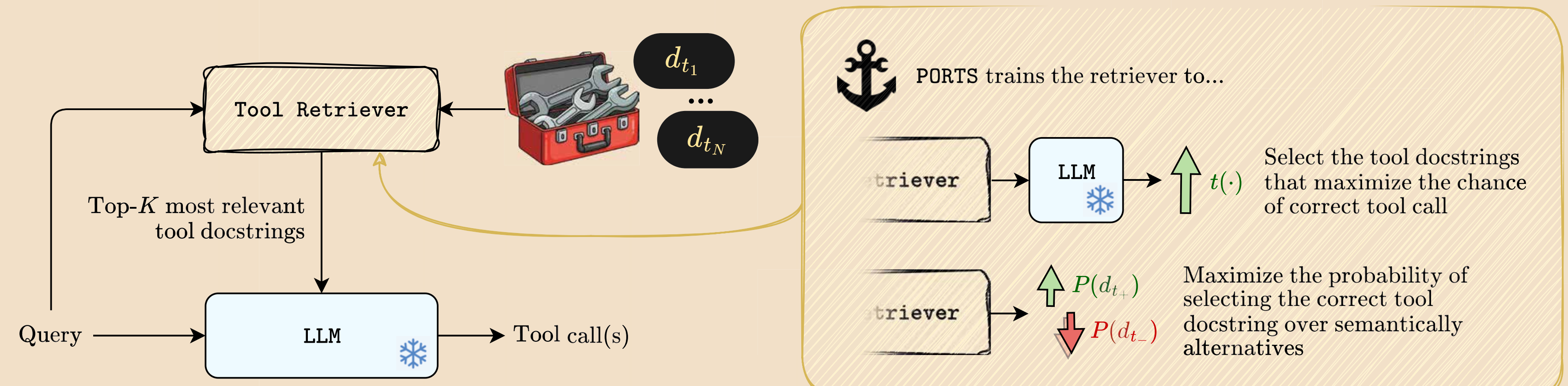


Fine-tuning LLMs is expensive and confines knowledge to a predefined set of tools. With many tools, their docstrings d_t may not fit in the prompt—or, if they do, finding the right one becomes a needle-in-a-haystack problem.

Recently, the focus has shifted toward **tool retrieval**. The retrieval component itself has seen little optimization.

Vanilla SOTA encoders struggle to select relevant tools. W/o fine-tuning, embeddings of d_t are highly concentrated in space—recurring data types, keywords, and concise but distinctive functional signatures.

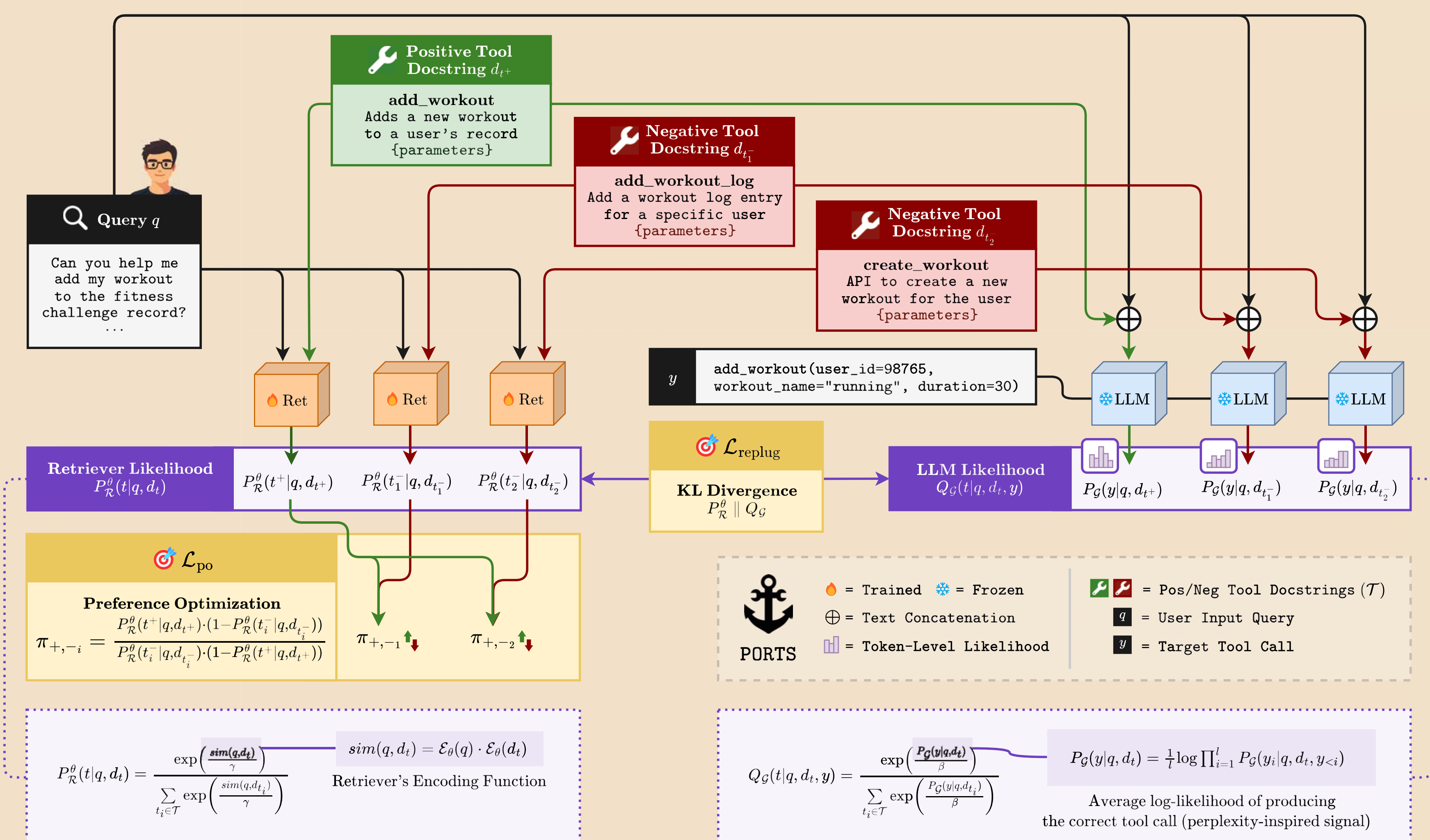
Method



We propose a new, **dual-objective training recipe** for tool retrievers:

- LLM-Alignment.** Building on RePLUG, we train the retriever so that the probability it assigns to each t reflects how likely a frozen downstream LLM is to generate the correct tool call when d_t is provided as context.
- Contrastive Learning.** We use hard-negative mining on d_t embeddings to pair each correct tool (t_+) with a highly similar but incorrect one (t_-). We train to maximize the odds ratio between the selection probabilities.

$$\mathcal{L}_{\text{PORTS}} = \mathcal{L}_{\text{replug}} + \lambda \cdot \mathcal{L}_{\text{po}}$$



Experimental Setup

6 popular **datasets**, single-tool selection (# of unique tools):

- ToolBench (12,934)
- API-Bank (1,960)
- APIBench (1,557)
- BGCL-v2 (1,015)
- ToolE (199)
- Octopus-v2 (20)

We create out-of-domain variants with unseen test tools

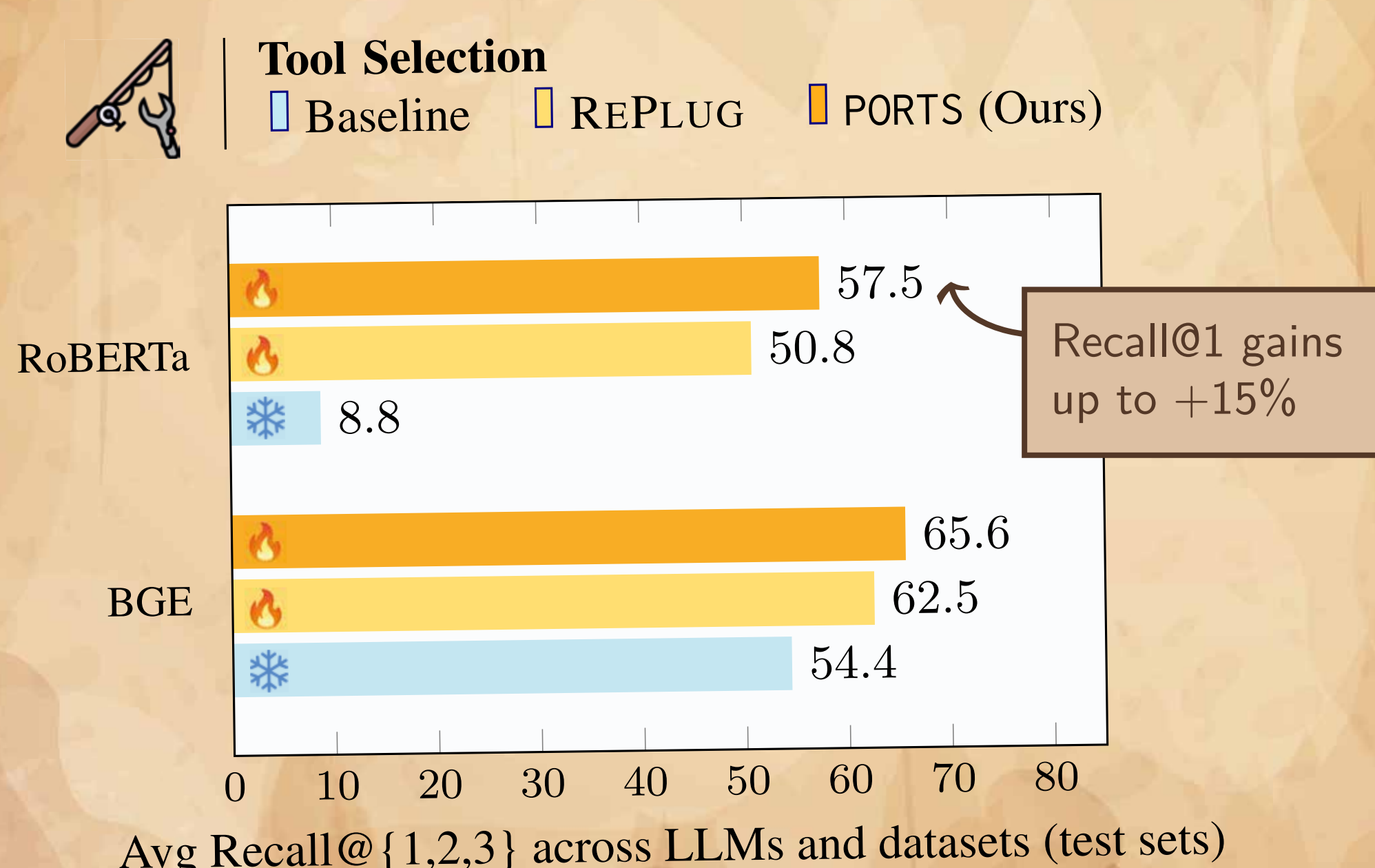
R models (base):

- RoBERTa-base, 125M
- BGE-base, 109M

G models:

- LLaMA3-8B
- Codestral-22B-v0.1
- LLaMA3-Groq-8B-Tool-Use

Large performance headroom above RePLUG



Results

Consistent, large absolute improvement over the base models

PORTS elevates base model effectiveness in *all* datasets, regardless of the selected LLM for supervision signals.

- For *seen* tools, it delivers an increase of up to +72% in avg Recall@{1,2,3} and +70% in NDCG@{1,3,5}.
- For *unseen* tools, gains remain remarkable at +61% in Recall and +59% in NDCG.

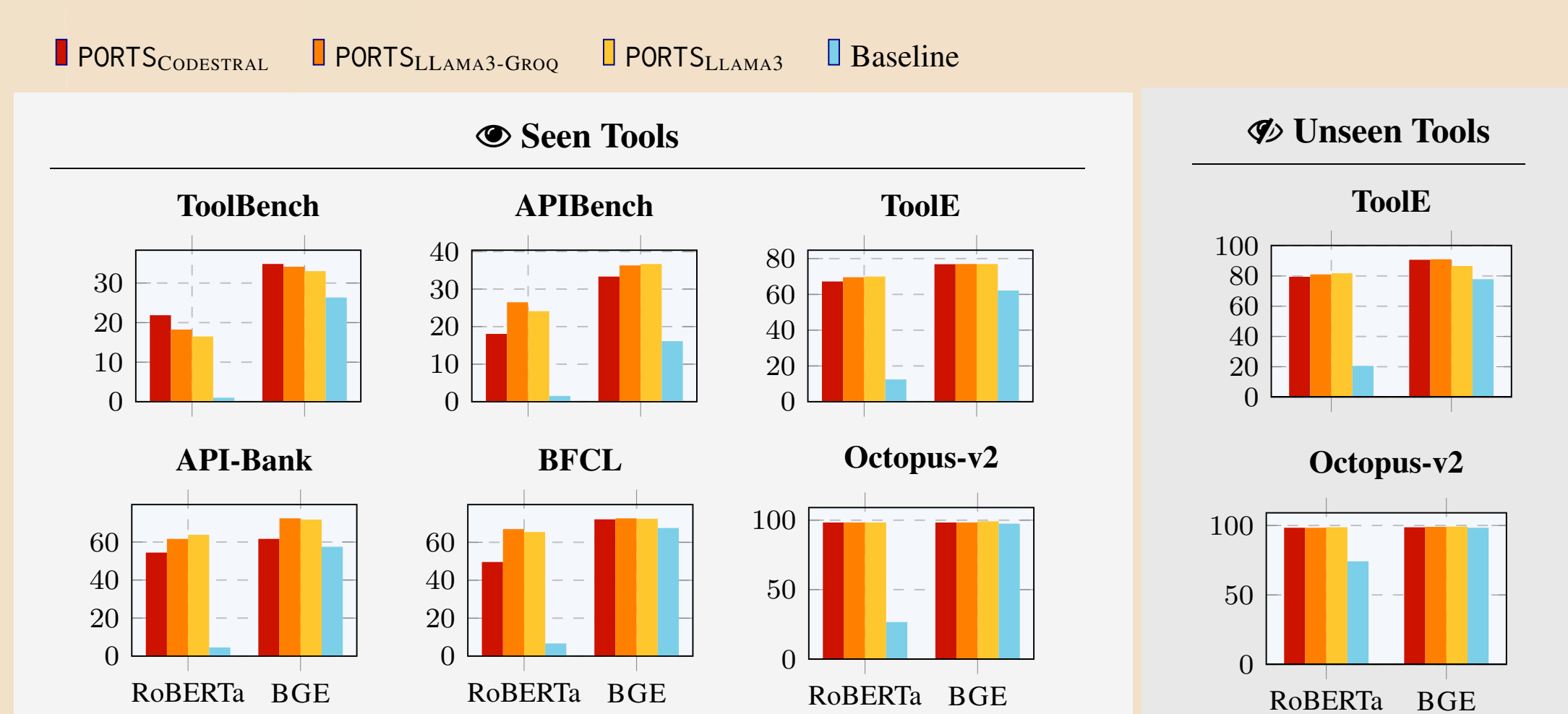
General-purpose encoders exhibit greater benefits: avg +47% Recall for RoBERTa, nearly 5x that of BGE (+10%).

Dataset characteristics dictate optimal LLM supervisor

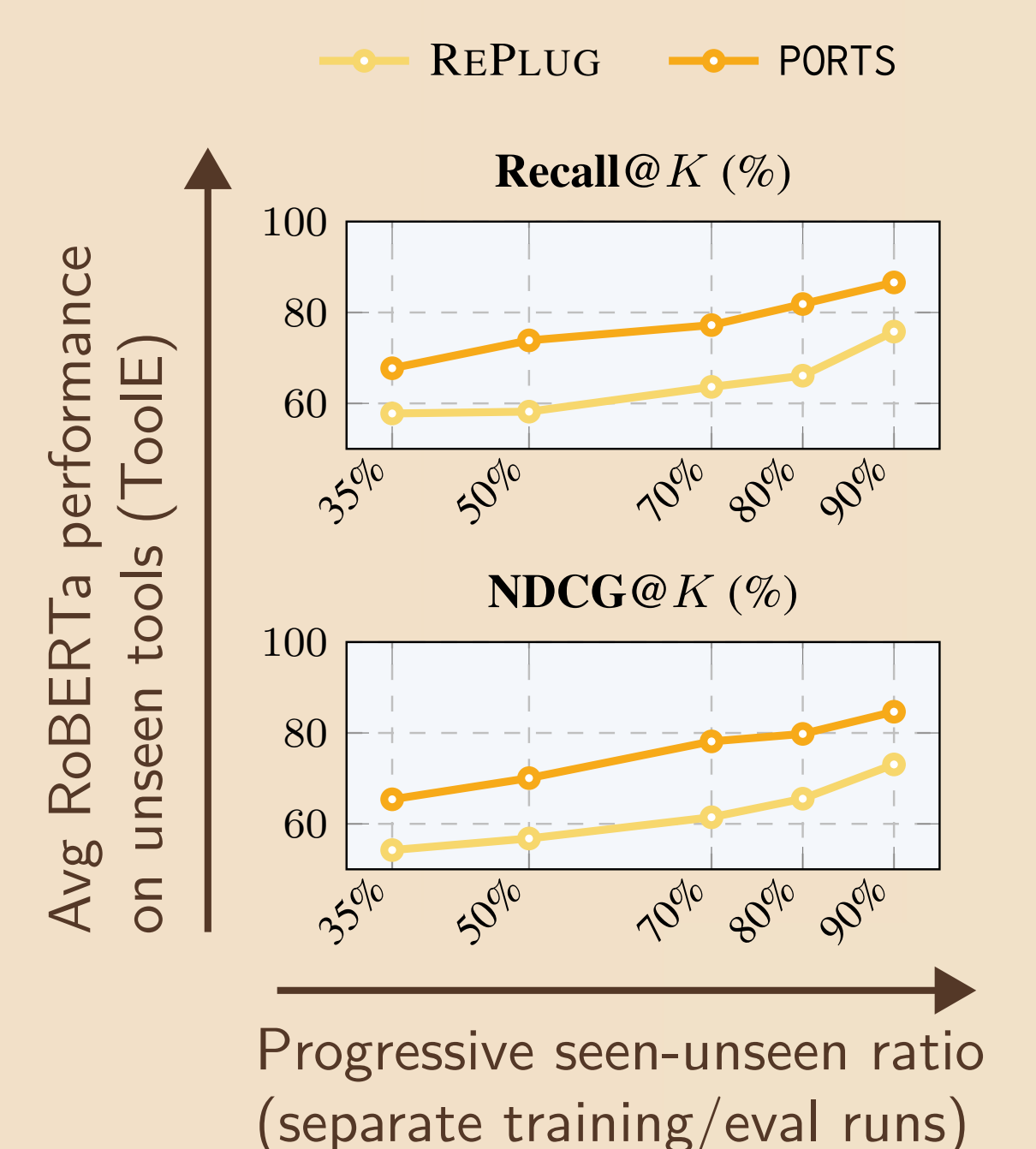
- The sensitivity to the supervising LLM—avg Recall Δ (max-min) across datasets—is 5.6% for RoBERTa and 2.9% for BGE.
- No single best LLM supervisor; the choice is mostly tied to the dataset. LLaMA3-Groq and Codestral excel in datasets with detailed docstrings, such as ToolBench.

Few training data, higher generalization than RePLUG

- PORTS is suitable even for scenarios with constantly evolving APIs.



Avg in-domain and out-of-domain Recall@K for each dataset (test set)



We publicly release models, datasets, and code (including the first open-source RePLUG implementation from scratch)